Braillix - Assembly Bible & Wiring Guide

Follow this guide sequentially to build your first single-cell prototype.

SECTION 1: CRITICAL SAFETY CHECKS (DO THIS FIRST)

> [!CAUTION] **BARREL JACK POLARITY CHECK** The photographed inline female pigtail has red/black wires, but colour is not proof. If you wire 5V backwards, the ULN2003 chip can burn out. 1. Plug the 5V adapter into the wall. 2. Plug the adapter's round male plug into the black inline female pigtail. Keep both stripped wires disconnected from the circuit. 3. Turn on your Multimeter and set it to **DC Voltage (V DC)**. 4. Touch the RED probe to the exposed red wire and the BLACK probe to the exposed black wire. 5. If the screen reads approximately `+5.0V`, red is **POSITIVE (+)** and black is **GND (-)**. 6. If it reads approximately `-5.0V`, the pigtail colours are reversed. Do not connect it until the wires are relabelled. 7. Disconnect wall power before attaching either wire to the breadboard.

SECTION 2: WIRING DIAGRAM

We are wiring the **ESP32**, **ULN2003**, and **Hall Sensor** together.

```
[5V/3A Adapter]
      |
      +---> (+) ---> [BREADBOARD 5V RAIL] ---> ULN2003 (+) power pin
      |
      +---> (-) ---> [BREADBOARD GND RAIL] ---> ULN2003 (-) power pin

[ESP32 (Powered by USB)]
   GND  -------------------------> [BREADBOARD GND RAIL] (Crucial: Common Ground!)
   3V3  -------------------------> Hall Sensor VCC
   GND  -------------------------> Hall Sensor GND
   D34  <------------------------- Hall Sensor AO (Analog Out)
   D18  -------------------------> ULN2003 IN1
   D19  -------------------------> ULN2003 IN2
   D21  -------------------------> ULN2003 IN3
   D22  -------------------------> ULN2003 IN4

[ULN2003 Output]
   White JST Socket -------------> 28BYJ-48 Stepper Motor
```

Kill-Shots to Avoid:

1. **Never** connect the 5V rail to the Hall Sensor VCC. ESP32 pins are not 5V tolerant. Use 3V3.
2. **Never** connect the 5V adapter rail to the ESP32 `VIN` pin while the USB cable is plugged into your laptop. Power the motor from the adapter, and the ESP32 from USB. Only link their grounds.

SECTION 3: STEP-BY-STEP ASSEMBLY SEQUENCE

Phase A: Motor Prep

1. Take the **28BYJ-48 motor** and mount it to the `base_plate` using two M4x10mm bolts.

(Note: The CAD assumes a specific shaft length. If the motor doesn't sit flush, we will need caliper measurements to update the CAD).

1. Press the resin `braille_cam` disc onto the motor's D-shaft.
2. Seat the Hall sensor into its pocket on the base plate.

Phase B: Electronics Fit

1. Snip off the female Dupont connectors on one end of your jumper wires.
2. Strip 3mm of wire, tin the ends, and solder them **flat** (parallel to the board) onto the ULN2003 header pins.

(Why? If you plug Dupont connectors in normally, they stand 20mm tall and won't fit in the 16mm printed electronics pocket).

1. Lay the ULN2003 flat on the floor of the `outer_box` pocket.
2. Plug the motor's white JST connector into the ULN2003.

Phase C: Mechanism Drop-In

1. Lower the mid-plate onto the z=20 ledge in the `outer_box`.
2. Lower the base_plate+motor assembly down into the box.
3. Take your 6 resin-printed `linkages`. Place the "foot" of each linkage onto its respective cam track.
4. The linkages should fan out naturally (60 degrees apart) so they don't hit each other.

Phase D: Top Plate & Springs

1. Glue the resin `dot_insert` tile into the PETG `top_plate` pocket using superglue or epoxy. Let it cure.
2. Thread a 2mm micro compression spring onto the dome of each of the 6 linkages. Give it a slight twist to seat it.
3. Lower the `top_plate` carefully over the standoffs, guiding the 6 braille dots through the holes in the resin insert.
4. Secure the top plate with four M2.5x25mm bolts.

Verification: Gently push down on each braille dot with your finger. It should spring back up smoothly without binding.

SECTION 4: BEGINNER SOLDERING MISTAKES

If you are doing the flat-soldering for the ULN2003:

- **Cold Joint:** Looks like a dull, grey blob that doesn't stick to the pad. *Fix: Apply a tiny bit of flux, heat the pin and wire simultaneously for 2 seconds, and reapply a dab of solder.*
- **Bridging:** Solder spills over and connects two pins (e.g., IN1 and IN2). This will break the motor steps. *Fix: Heat the bridge and swipe your iron away quickly, or use desoldering wick to soak up the excess.*
- **Melted Insulation:** Holding the iron on the wire too long melts the plastic jacket. *Fix: Tin the wire and pin separately first (apply solder to them individually), then just touch them together with the iron for 1 second to fuse them.*